

Low Power VLSI Circuits Design Strategies and Methodologies: A Literature Review

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Abstract— Researchers stare at the design of low power devices as they are ruling the today's electronics industries. In VLSI circuits, power dissipation is a critical design parameter as it plays a vital role in the performance estimation of the battery operated devices particularly used in biomedical applications. The decrease in chip size and increase in chip density and complexity escalate the difficulty in designing higher performance low power consuming system on a chip. Further, overall power management on a chip is becoming a big challenge below 100 nm node because of its increased design complexity. Besides, leakage current also plays a vital role in Power management of low power VLSI devices. In sub-micron technologies, leakage and dynamic power consumption is becoming an essential design parameter as it is dissipating a considerable portion of the total power consumption. To increase the battery life of portable devices, leakage and dynamic power reduction is emerging as a primary goal of the VLSI circuit design. This paper provides an insight about the various methodologies, strategies and power management techniques to be used for the design of low power circuit based systems.

Index Terms— CMOS, Power Dissipation, Low power strategies, Dynamic Power Management, Leakage power

I. INTRODUCTION

There is a big boom in microelectronics industry since the invention of transistor which lays the foundation for low power consuming devices. The integrated circuits (IC) improved the performance of the circuits and also led to size reduction. This results the growth of power unit area in IC. A remarkable growth in battery powered complex functional life saving devices like pacemakers and other implantable medical instruments urges the requirements of low power consuming devices. Researchers focus more than ever before in evolving the low power components and low power design techniques. On the other side, silicon failure rate is doubled for every 10°C rise in temperature in high power consuming devices. Reducing power is a preternatural development which gathered importance with the developments of deep submicron nodes and nanometer technologies.

II. SOURCES OF POWER DISSIPATION

In an electrical or electronic system, the rate at which energy is dissipated over a time is known as power dissipation which is classified as two types namely, peak power and average power. Peak power is the maximum instantaneous power over a time and it affects reliability of the device which results in glitch and in turn increases the possibility of malfunction of the system. Average power affects the packing and cooling systems. The total power dissipated by a circuit is given by [1]

$$p(t) = i(t)v(t) \quad (1)$$

$i(t)$ is the instantaneous current and $v(t)$ is bias voltage supplied by the power supply. Power minimization techniques concentrates on average power or maximum instantaneous power. The prime power dissipation sources of CMOS circuits are static, short-circuit, and dynamic power. The total power dissipation for CMOS circuits is as follows [1] [8]

$$P_{total} = P_{switching} + P_{short} + P_{leakage} \quad (2)$$

$$P_{switching} = \alpha C_L V^2 f \quad (3)$$

$$P_{short} = \tau \alpha V I_{short} \quad (4)$$

$$P_{leakage} = V I_{leakage} \quad (5)$$

C_L is the node switching capacitance, f is the frequency of switching, V is the supply voltage, α is the switching activity, and τ is the voltage settling time.

Leakage or static power is the power consumed by a device due to transistor leakage and it is produced by the reverse biased current is shown in Fig.1.[2] The causes for the reverse biased current are the sub threshold leakage, gate oxide tunneling leakage, diffusion region and substrate. The algebraic sum of all the biased current and leakage current constitutes the static power.[2][17]

In deep submicron technology process, leakage power is considered as an important parameter; it is generated when the CMOS device is operated in both static and switching mode. In general, major problem with the leakage power is when a device is in idle state where all the consumed power is absolutely wasted.

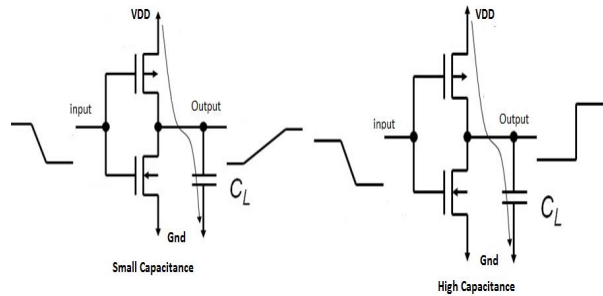


Fig. 1-Leakage power dissipation due to low and high load capacitance

Short circuit power dissipation is due to the supply voltage V_{dd} of devices, $V_{threshold}$ of transistors, rise and fall time of input. An input with slow rise time in COMS inverter contributes nearly 20% to the short circuit power dissipation.[2] Static power dissipation is more effectively reduced by maintaining the equilibrium between input transition and gate output transition and gates which are working at high switching speed consume more short-circuit power is shown in Fig.2.[8] Increasing the output load capacitance also reduces short-circuit power. Scaling down of supply voltage with respect to threshold voltages reduces short circuit power. Further, it is believed that introducing newer technologies in the IC fabrication process can also reduce the short circuit power dissipation.

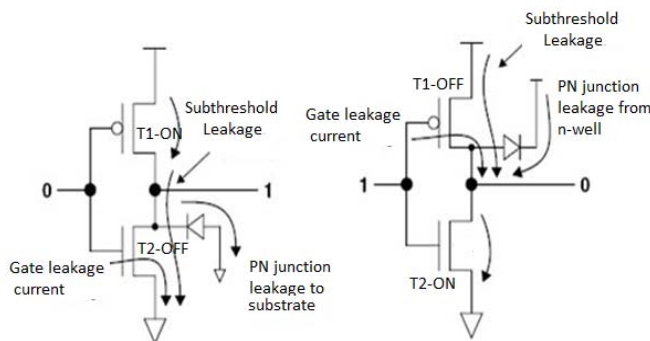


Fig. 2 - Short circuit power dissipation

Dynamic power is the power which is consumed by a device during the parasitic capacitance charging and discharging process due to oxide gates and metal footprints; the dynamic power consumption is the maximum power consumption when compared to the overall power consumption of a chip.[4] Hence, dynamic power dissipation reduction is more important than the static or leakage power. Generic representation of a CMOS logic gate for computing the switching power consists of drain capacitance, interconnect capacitance and input capacitance is shown in Fig.3.

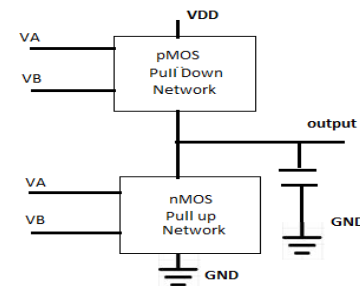


Fig. 3 - Generic model of dynamic power computation

The switching of logic elements with respect to its input is known as toggle rate which depends on frequency, input voltage and capacitance is shown in Fig.4.[14] The toggles may be in conducting wire of interconnect, external package pins or internal logic modules.

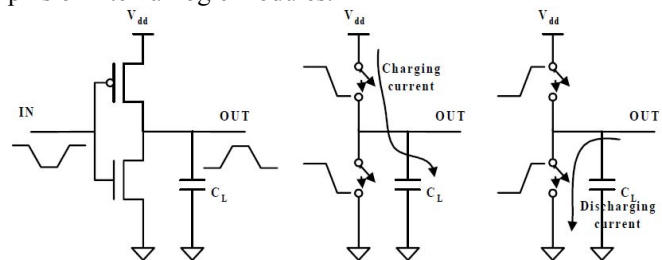


Fig. 4 - Dynamic power dissipation: charging and discharging

A CMOS gate switching power dissipation under synchronous condition is given by [3]

$$P_{switching} = \alpha C_L V_{dd}^2 f \quad (6)$$

Dynamic power dissipation [2] is not a function of transistor size and it depends on the design parameters of node capacitance C_L , supply voltage V_{dd} and the switching frequency f . Dynamic power consumption can be reduced by decreasing the value of the design parameters during the development stage itself. Further, power dissipation is also a data dependent function of switching activity. Power optimization can be achieved only by solving power related issues in all the design stages and the technology used for VLSI circuits. Practically, the supply voltage scaling is the most suitable form of power optimization technique, because it yields a noticeable power savings due to the quadratic dependency of switching power on supply voltage.[12] [18] However, lowering the supply voltage affects the operating speed of the circuit.

III. THE LOW POWER DESIGN FLOW

A top down approach in low power design flow with power analysis and power estimation steps is illustrated in Fig.5. which summarizes the flow of steps that are necessary for a system from a system level to architecture level specifications. Low power design flow naturally mandates the number of steps to be carried out by a design engineer for a specific application. Every design level has opportunities for employing power optimization techniques and each level can

contribute significantly to achieve more power reduction in the VLSI circuits. However, some combination of low power design techniques rather than considering single one in each stage may yield good results which strongly depend on the application requirements.

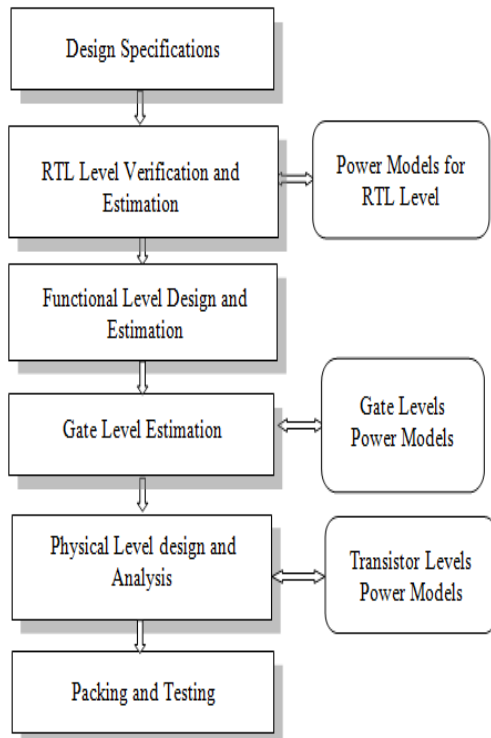


Fig. 5- Low power design flow with power estimation steps

A low power technique requires optimization at the higher abstraction levels, which leads to large amount of power savings. Table I shows the possible percentage of power reduction [5] through the profiteering opportunities at various levels of the design hierarchy with bottom up approach.

Table I Power savings in terms of the design level

Levels of Design	Power Reduction in percentage
Behavioral / System Level	70-90
Functional Level	10-20
Gate Level	15
Physical / Transistor Level	30
Layout Level	20

IV. CIRCUIT AND LAYOUT LEVEL OPTIMIZATIONS

Scaling is a technique to reduce the power dissipation in VLSI technology and it becomes an amazingly a fast and widely used step towards minimizing the power in VLSI circuits for the last decades.[13] A constant field scaling requires an aggressive scaling of geometric features and scaling the levels of silicon doping to maintain constant field across the gate oxide of the CMOS transistor.

Power dissipation is scaled as m^2 while speed increases by a factor “m” for constant field devices and power density

remains constant. Power consumption does not follow this swing as the transistors are integrated in real circuits and both the power density and average power tends to increase as minimum feature size shrinks. Further, die size increases steadily with technology which results in an increase of average power and supply voltages which are not reduced in proportion to the decrease of constant field scaling.[13] Constant voltage scaling is widely used in early stage of 0.8 μ m silicon technology generation.

CMOS models with accurate transistor currents are predominant for producing suitable scaled supply voltage (V_{DD}) and threshold voltage (V_T) values. For sub micrometric transistors, a CMOS ON current quadratic model provides an extremely optimistic switching speed estimates which is given in equation 7. [8]

$$I_{DS} = S'(V_{GS} - V_T)^2 \quad (7)$$

The saturation velocity (S') of the electrons traveling between source and drain in short channel transistors is given by the equation 8[8]

$$I_{DS} = S'(V_{GS} - V_T)^m; \quad 1 \leq m \leq 2 \quad (8)$$

The sub-threshold conduction is another important characteristic of CMOS transistors, when $V_{GS} < V_T$, the current is non zero, but it has an exponential growth which is given by equation 9[8]

$$I_{DS} = R'e^{V_T/V_0} \quad (9)$$

V_0 is sub threshold slope, R' the internal resistance.

Velocity saturation plays a vital role in supply down-scaling, however, down scaling limited by sub-threshold conduction, because of the increase in static current leakage through marginally OFF transistors.

Another approach is ultra low power (ULP) technique optimizes supply and threshold voltages to balance ON and OFF currents to maintain superior performance. [9][10][18] To acquire an acceptable minimum energy delay product, supply and threshold voltages should be maintained as minimum. Further, supply voltage should be insignificantly higher than twice of threshold voltage V_T which is in range of 50 to few hundred millivolts. Threshold and supply voltages are not scaled to their maximum possible value, but both are reduced much with respect to constant voltage scaling.

In general, ultra low power MOS transistors are not considered for real life applications because of two reasons. One reason is that sub threshold currents of many transistors today are at higher magnitudes than the expected that results the threshold control is not perfect. The second reason is, the temperature varies exponentially with sub threshold current, which requires very good heat sink or cooling structure to control the temperature rise on ultra low power MOS transistors, which is more expensive process but aggressive voltage scaling finds a common place in low power VLSI circuit design.[11] Besides, reducing power by aggressive voltage scaling has major drawbacks. Down scaling with

minimum channel length in CMOS transistors are leaky in nature during the OFF state and further voltage scaling is not possible because of the threshold voltage limitation.

In recent technologies, leakage power is a major problem, because it directly affects the life of the battery even though the circuit is working in standby or idle mode. Multiple threshold and variable threshold circuits are developed in order to meet the leakage power issues. In multiple threshold CMOS, various levels of thresholds are employed in which low threshold voltage transistors are employed in a speed critical sub systems which are fast and leaky in nature and high threshold transistors are quite slow and exhibits low sub threshold leakage, which is employed in noncritical paths of integrated circuits. On the other hand, multiple threshold techniques become less effective when more than one transistor becomes timing critical. Substrate biasing effectively overthrows the multiple threshold issue using variable threshold circuits.

Under idle state, NMOS transistor's substrate is biased negatively in variable threshold circuit which increases the threshold voltage due to the body bias effect. A stand by control units which modulates the substrate voltage which are required to overcome the idle state leakage power problem.

Variable and multiple voltage techniques are refined to solve leakage power problems. Two or more supply voltage levels are employed in multiple voltage circuits. In multiple threshold techniques, high threshold voltages are applied to high speed transistors and low threshold voltage is applied to the non critical type transistors. Powering the on-chip internal logic circuits at much lower voltage to save power, multiple voltages are employed to provide the standard voltage levels to I/O circuits. In multiple power supply voltage scheme, distribution of multiple power supply rails in internal logic and the energy efficient level shifters, to interface high to low voltage sections and vice versa, is major challenge.

Power reduction using variable voltage technique modulates the power supply voltage based on the logical operation which is a commercially profitable technique as it offers a way to switch data between power dissipation and speed during its run time.[14] The above voltage scaling techniques requires more powerful system support and process, entails an increase in total cost which are suitable only for large volume of applications.

Aggressive voltage scaling at circuit level techniques, like library cell design and transistor sizing for reduced power consumption have initiated momentum towards low Power dissipation.[7] Sequential primitives such as latches and flip-flops are the most important cells in a digital library because they are parallel processing circuits and connected to the clock pluses which are the highest active part of a network in integrated circuits.

Clock drivers are considered as the highest power dissipating network in an IC due to the presence of large load capacitance at the clock distribution network. Low power consuming flip flop and latch design focuses on reducing load due to load and internal power consumption during the clock signal which is switching from logic zero to one and vice versa is shown in Fig.6.[8] Optimization using clock power is performed at the transistor level to upgrade the power distribution tree and circuits driven by clocks and oscillatory circuits to generate clock pulses.

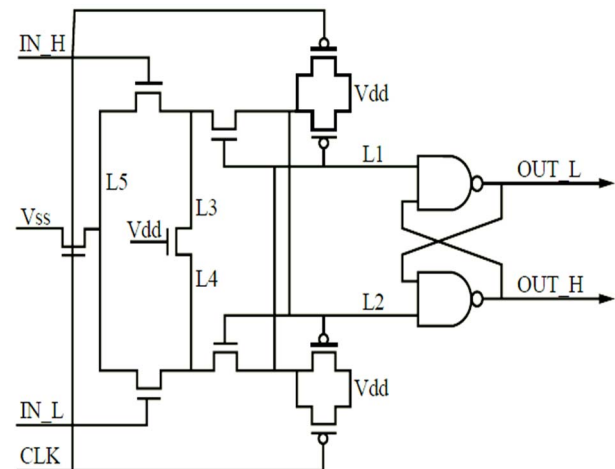


Fig. 6 - A low power Flip- Flop circuit

V. LOGIC AND ARCHITECTURE LEVEL OPTIMIZATIONS

Optimization at logic level is really a costly approach in terms of time consumption for designing a circuits and logical effort, but more cost effective for a structured logic while manufacturing large volume components like microprocessors. Fortunately, an automated commercially available synthesis tool at logic level provides logic level power optimization to unstructured logic and for low volume VLSI circuits.[15]

Logic optimization is a process in which power supply voltage levels are set at a fixed value and the size of the logical gate is fixed to reduce power consumption. In circuit level as well as technology level techniques, power consumption is not only the parameter and performance which is also a constraint. For constrained power optimization, all logical subsystems are converted to consume minimum power with path equalization technique and the critical path lengths are equalized.

Path equalization maintains a similar path length when a signal propagates from an input to output of a network in integrated circuits. Mostly, the aligned transition at the inputs of gates minimizes the switching activity. Path equalization is a technique most widely used in multipliers where adders are the processing units to perform arithmetic operations.

Arithmetic logic units have less irregular structures than the glue logic and its gate level implementations are represented by the delays in path where power optimization is achieved by altering the size of transistors which is known as resizing. Resizing concentrates the fastest combinational circuit path because, logic gates on fastest paths are down scaled in size to decrease their capacitance at the input. Propagation delays are equalized by slowing down the fast paths and the power consumption is reduced by decreasing the capacitance values and avoiding fake switching which is created by the glitches. It is believed that transistor resizing is an improvement technique that provides an exchange between output swing and short circuit power due to internal logic on many gates at an equivalent time.

Local transformation techniques like remapping, re factoring, pin swapping and phase assignments are the other logical level power optimization techniques performed at gate level netlists with large switching capacitance.[3] Logic level optimization does not lead to high power reductions. An average 10 to 20% power saving is possible at logic level optimization and the power optimization at higher levels of abstraction will provide more noticeable saving on power.[8]

In general, complex digital logic circuits have some non essential units that are not required for performing useful computations at every clock cycle. These units are disabled during the corresponding clock periods, aiming towards the power reduction. Disabling a particular unit from performing useless switching results in a reduction to the total switching capacitance of the circuit. This switching capacitance reduction process in turn decreases the power dissipated by the switching elements which is called as dynamic power management (DPM) methods. DPM is widely used at system-level design.[19]

Clock gating is a scheme applied for a specific sub system to disable the clock input and preventing the subsystem from switching activity at any time of the computation performed by a hardware section is useless.[6][16] A block diagram of simple sequential circuit is shown in Fig.7.[5]

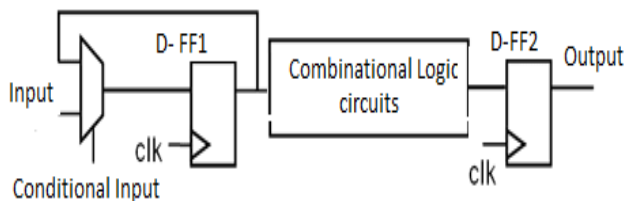


Fig. 7 - Example of simple sequential architecture

This sequential circuit is made up of combinational logic circuits and a group of registers for storing the states. A feedback path is established between the next state logic and combinational circuits.

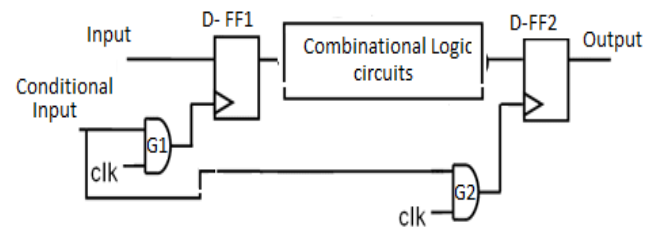


Fig. 8 - Example of gated clock architecture

The block diagram of gated clock structure is shown in Fig.8.[5] All the storage purpose registers are made up of edge triggered flip flops and a single clock pulse is applied to trigger the circuit. Combinational logic block is under the control of primary inputs, present inputs and the primary output states of the circuit act as trigger to activate the mechanism for clock gating.

For the proper operation, a global clock pulse is applied to the circuit. During this clock, the latch becomes transparent and minimizes the effects of glitches generated at the output blocks. A proper synthesis of the G_1 and G_2 reduces any timing violations as the logic for the activation function is on the critical path. Developing a system based on all the above conditions requires more additional logic circuits which will consume more power. Besides, it is necessary to design the circuits with minimum gate count which will dissipate lesser power. The system speed is affected by clock gating as the clock gating produces a delay which affects the system speed.

VI. SYSTEM AND SOFTWARE LEVEL OPTIMIZATIONS

Nowadays, electronic gadget comprises of hardware platforms with the sequence of many layers of software. Several system features relay on interaction between the hardware and software layers. Normally during execution, Software doesn't consume energy but it will consume power when it is stored in a memory based utilization of hardware.

Software execution is a process that performs all tasks at hardware and accesses memory to write and read operations. Hence program execution using software consumes power during mathematical calculations, storage of data and communication between other devices. Further, semiconductor memories like dynamic RAM and SRAM are used to store computer programs which will also consume power for refreshing and its normal operations. The power consumed by program storage is very less and predictable during the design itself. Thus, the power dissipation during the software execution is reduced by decreasing the program length or the number of lines in the program. Additional reduction in program length is obtained using compression techniques.

The power consumed during the executing of a program relies on its low level machine code and the parameters of the hardware architecture. The source code is converted into low

level machine code through software compilation. Normally, machine code is affected during the software compilation process.

Software instructions are represented by the amount of clock cycles required for execution and the amount of the energy needed for every clock cycle. The power dissipation during each instruction slightly depends on the processor busy/idle status. Moreover, during the memory based instructions, power consumption will significantly be increased due to the data access from the memory. By minimizing the program code length, compiler increases the execution speed for the generated machine code and also minimizes the loss of data in memory.

In general, various levels and styles can be used in developing software source code at the cost of power consumption. Energy efficient source code for software mandates certain code writing approaches or automated source code transformation to decrease the power consumption of a program. An energy efficient operating system can be evolved using power aware task scheduler. Dynamic power management system offers a better energy saving in power aware operating systems.

VII. CONCLUSION

The aim of electronic design is to create a balance between power efficiency and performance in terms of speed. Design of VLSI circuits for low power applications is a multi faceted problem as the circuit designers have to follow several degrees of freedom to have acceptable power reduction. A low power design flow must satisfy all power consumption issues in each stage of the design process and abstraction levels to optimize power consumption.

In this paper, various strategies and methodologies used for minimizing the leakage and dynamic power are presented. The strategies and methodologies which are discussed in this paper are very much useful for the designers to design low power VLSI circuits which are used in portable and biomedical applications.

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